

Question

A is an important compound of the element **M** with a point group of D_{2d} , where the mass fraction of **M** is 70.0%. The elemental form of **M** can be obtained by the fusion reaction of **A** with tin. Heating the elemental **M** in air yields **B**. Refluxing **B** with S_2Cl_2 under chlorine gas produces the oily liquid **C**. If **B** is mixed with Zn and hydrochloric acid, one of the products reacts with silver nitrate solution to form a black precipitate **D**. Please determine which of the following options is correct.

- A.** Element **M** is an element from the fifth period.
- B.** There are 5 atoms in compound **A**.
- C.** The mass fraction of oxygen in compound **B** is 43.66%.
- D.** The molar mass of compound **C** is approximately 181.27 g/mol
- E.** The sum of the molar masses of **M** and **D** is 304.9 g/mol
- F.** In the reaction equation where **B** is heated under reflux with S_2Cl_2 under chlorine gas, there are three products.

Answer

Correct Answer: **D**

Detailed Explanation

Determine the element **M**:

1. Heating the simple substance **M** in air can yield **B**, then **B** should be the oxide of **M**.
2. Mixing **B** with Zn and hydrochloric acid, one of the products reacts with silver nitrate solution, and the formation of a black precipitate **D** can be observed. Zn-HCl is a reducing system, then the oxide **B** will be reduced when mixed with Zn and hydrochloric acid, and this product reacts with silver nitrate solution, and the formation of a black precipitate **D** can be observed. If no redox reaction occurs, the black precipitate may be Ag_2S (Ag_2O cannot be produced in this way, and Ag_2Se is a gray crystalline powder); if a redox reaction occurs, the black precipitate may be elemental Ag, and **B** may be the reducing gas PH_3 , AsH_3 , SbH_3 .
3. Heating **B** with S_2Cl_2 under chlorine gas reflux can obtain an oily liquid **C**. This information can rule out the possibility that **M** is sulfur, so the black precipitate **D** is Ag. Considering that S_2Cl_2 is generally used as a chlorinating reagent, and the chloride is liquid, it can be locked that **M** is one of P, As, and Sb.

CHECKPOINT

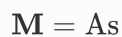
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M is one of P, As, Sb

4. The point group D_{2d} is common in **tetrahedral twisted structures**. Let $\mathbf{A} = \mathbf{MX}_n$, the molar mass of **M** is $M(\mathbf{M})$, and the molar mass of **X** is $M(\mathbf{X})$, then the mass fraction expression is: $w = \frac{M(\mathbf{M})}{M(\mathbf{M}) + nM(\mathbf{X})}$, try to substitute common combinations to find a compound with a mass fraction of about 70%. A reasonable result is $\mathbf{M} = \text{As}$, $\mathbf{X} = \text{S}$, $n = 1$, then **A** is As_4S_4 .

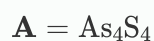
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Verify subsequent compounds:

- **B**: As is heated in air to obtain an oxide. Since pentavalent As has strong oxidizing properties, it can only be As_2O_3

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- **C**: As_2O_3 and S_2Cl_2 are heated under reflux in chlorine gas to form an oily liquid **C**.
- Reaction: $2\text{As}_2\text{O}_3 + 3\text{S}_2\text{Cl}_2 + 2\text{Cl}_2 \rightarrow 2\text{AsCl}_3 + 6\text{SO}_2$
- **C** = AsCl_3 , oily liquid, molar mass = 181.28 g/mol

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- **D**: As_2O_3 is mixed with Zn and hydrochloric acid
- Reaction: $\text{As}_2\text{O}_3 + 6\text{Zn} + 12\text{HCl} \rightarrow 2\text{AsH}_3 \uparrow + 6\text{ZnCl}_2 + 3\text{H}_2\text{O}$
- AsH_3 passed into silver nitrate will produce silver element!

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D = Ag

Analyze options:

A: As is a fourth-period element, incorrect

B: As_4S_4 has 8 atoms, incorrect

C: Mass fraction of oxygen in $\text{As}_2\text{O}_3 = 24.26\%$, incorrect

D: The molar mass of AsCl_3 is 181.27 g/mol, correct

E: The sum of the molar masses of As and Ag is 182.79 g/mol, incorrect

F: $2\text{As}_2\text{O}_3 + 3\text{S}_2\text{Cl}_2 + 2\text{Cl}_2 \rightarrow 2\text{AsCl}_3 + 6\text{SO}_2$, there are two products, incorrect

Therefore, the correct answer is option **D**.